

Experiment 31:-

Construct a C program to simulate the First in First Out paging technique of memory management.

Code:-

```
#include <stdio.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n)
{
    for (int i = 0; i < n; i++)
    {
        if (frames[i] == -1)
            printf(" - ");
        else
            printf(" %d ", frames[i]);
    }
    printf("\n");
}

int main()
{
    int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};

    int n = sizeof(referenceString) / sizeof(referenceString[0]);

    int frames[MAX_FRAMES];
```

```
int framePointer = 0;
for (int i = 0; i < MAX_FRAMES; i++)
{
    frames[i] = -1;
}
```

```
printf("Reference String: ");
```

```
for (int i = 0; i < n; i++)
{
    printf("%d ", referenceString[i]);
}
```

```
printf("\n\n");
```

```
printf("Page Replacement Order:\n");
```

```
for (int i = 0; i < n; i++)
{
    int page = referenceString[i];
    int pageFound = 0;
    for (int j = 0; j < MAX_FRAMES; j++)
    {
        if (frames[j] == page)
        {
            pageFound = 1;
            break;
        }
    }
}
```

```

    }
}
if (!pageFound)
{
    printf("Page %d -> ", page);

    frames[framePointer] = page;
    framePointer = (framePointer + 1) % MAX_FRAMES;
    printFrames(frames, MAX_FRAMES);
}
}
return 0;
}

```

Output:

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
```

```
Page Replacement Order:
```

```
Page 7 -> 7 - -
```

```
Page 0 -> 7 0 -
```

```
Page 1 -> 7 0 1
```

```
Page 2 -> 2 0 1
```

```
Page 3 -> 2 3 1
```

```
Page 0 -> 2 3 0
```

```
Page 4 -> 4 3 0
```

```
Page 2 -> 4 2 0
```

```
Page 3 -> 4 2 3
```

```
Page 0 -> 0 2 3
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.█
```